1. Write prototype of a function that takes a String and character and returns integer
2. Write method signature of the following function: int sum(int a, char ch, float f){}
3. Find output:

```
class Values{
int x, y;

Values(int a, int b){
        x=a;
        y=b;
}
void display(){
        System.out.println(x+ "" + y);
}

public static void main(String args[]){
        Values ob1=new Values();
        Values ob2=new Values();

        ob1.display();
        ob2.display();
}
```

4.
```
int x= 10, y=20;
x *= x++ + ++x - --y;
x=?
```
5.
```
int x=5, y=10;
x-= (x-- - --x) % y
x=?
```
6.
```
int x=5, y=10;
x-= (x-- - --x) / y
x=?
```
7. System.out.println(Math.pow(36,1/2) + 4);
8. Convert the following to ternary and switch case
```
int x;
if(x==10 || x==20)
        x=100;
else if( x%2==0 )
        x=1000;
else x= 10000;
```

9. Convert to ternary and switch case
```
if(x==y and y==z){
        System.out.println("All numbers are equal");
}else{
        System.out.println("Not all numbers are equal");
}
```
10. System.out.println(Math.sqrt(-25));
11. Using Math.random() generate a random number between 10 and 40 (both included)
12. What would the following expression return ?
(int)(Math.random()*6)+1

13. System.out.println(!(!(2>3) && (3<4) || 4 !=4));
14. System.out.println(3 + 4 + “Hello” + 3 + 4);

Loop snippets:

1.
```
int a=10, b=15;
for(int i=1; i<=5; i++){
        a++;
        b--;
}
System.out.println(a+b);
```

2.
```
int a=10, b=15;
for(int i=1; i<=5; i++)
        a++;
        b--;
System.out.println(a+b);
```

3.
```
int a=10, b=15;
for(int i=1; i<=5; i++);
        a++;
        b--;
System.out.println(a+b);
```

4.
```
int a=5;
while(a++ < 10){
        a+=5;
}
System.out.println(a);
```

5.
```
int a=5, i=1;
while(++i < 5){
        a+=i;
}
System.out.println(a);
```

6.
```
ini a=5, b=5;
do{
        a+=b;
        a++;
        b--;
}while(a<=10);
```

7.
```
for(int i=1; i<=5; i++){
        for(int j=i; j<=5; j++){
                System.out.println(i+ “” +j);
        }
        System.out.println();
}
```

8. for(int i=1; i<=5; i++){
```
for(int i=1; i<=5; i++){
        for(int j=1; j<i; j++){
                if(i==2) continue;
                if( i%2==0) System.out.print("*");
                else System.out.print("#");
        }
}
```

1. write a class to overload a function Print() as follows

class name: Series

member methods:

Print(int n) to generate Floyd triangle with n rows

1

2 3

4 5 6

7 8 9 10

Upto n rows

Print(int n, int x)

$$S=\frac{x+2y}{3}+\frac{2x+3y}{4}+\frac{3x+4y}{5}+\frac{4x+5y}{6}+\ldots\ldots+\frac{nx+(n+1)y}{n+2}$$

Print(double n){

$$S=\sqrt{\frac{2x+y^3}{4!}}+\sqrt[3]{\frac{3x+y^4}{5!}}+\sqrt[4]{\frac{4x+y^5}{6!}}+\sqrt[5]{\frac{5x+y^6}{7!}}+\ldots+\sqrt[n]{\frac{nx+y^{n+1}}{(n+2)!}}+$$

}

1. Define a class named ParcelDelivery with the following specifications:

Member Variables:

int parcelId – stores the parcel identification number

int weight – stores the weight of the parcel in kilograms

double totalCost – stores the total delivery cost to be paid

Member Methods:

ParcelDelivery() – default constructor to initialize data members with default values

void inputDetails() – prompts the user to input their parcel ID and parcel weight using **Scanner** class methods only

void computeCost() – calculates the delivery cost based on the following weight-based pricing:

Weight Range	**Price per Kg (Rs)**
First 5 Kg	20
Next 10 Kg	15
Above 15 Kg	10

void showDetails() – displays the parcel ID, parcel weight, and total delivery cost